The Stack Is Part of the Result: Negative Results on Attributing GRPO-Family Differences

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Abstract

Comparative claims about reinforcement-learning post-training of language models are usually made between method labels: "GRPO beats PPO", "DAPO beats GRPO". We report a set of negative and cautionary results showing that, in the GRPO family, the label is not the unit of evidence: the training stack is part of the result. A matched same-stack PPO/GRPO contrast on synthetic arithmetic is inconclusive ($\Delta = -0.002$, $p = 0.374$, five seeds). A same-stack audit of four labelled variants on GSM8K is inconclusive in every contrast against baseline GRPO (eight seeds). Under one label, last-10 training reward spans 5.0% to 85.6% (0.050 to 0.856) across two stacks that also differ in base checkpoint, so it is not a backend-only effect; one label ("DAPO") yields opposite telemetry on two stacks, one of which is our own custom loss without dynamic sampling; and four labels implemented as our own simplified losses on one managed stack land within a 0.034 training-reward band. Two historical exhibits whose source analyses cannot be reproduced are withdrawn. We distil these exhibits into an eight-item minimum reportable stack and a machine-readable registry whose diff tool flags such label collisions from manifests alone.

1 Introduction

Group Relative Policy Optimization (GRPO) (Shao et al., 2024; DeepSeek-AI et al., 2025) replaces PPO's learned critic (Schulman et al., 2017) with a baseline computed from a group of G sampled completions per prompt, and has spawned a family of variants such as DAPO (Yu et al., 2025), Dr. GRPO (Liu et al., 2025) and GSPO (Zheng et al., 2025). Each is available in several libraries, including TRL (von Werra et al., 2022), veRL (Sheng et al., 2024), OpenRLHF (Hu et al., 2024) and managed services such as Tinker (Thinking Machines Lab, 2025), which differ in answer parser, sampler, KL handling, clip range, LoRA targets (Hu et al., 2022), precision and evaluator. Deep-RL and RLHF results are known to be sensitive to such details (Henderson et al., 2018; Engstrom et al., 2020; Huang et al., 2024), but papers in this area are not asked to report them, so a gain attributed to a method label may belong to the stack.

This short paper asks a narrow question: *how much of a reported GRPO-family difference survives once the stack is held fixed?* Our answer, from our own short-horizon, low-seed GSM8K experiments, is: little that we could detect. We contribute (i) five measured exhibits of stack sensitivity (E-2, E-4–E-7), including the E-2 failure to detect and E-4 inconclusive contrasts, not equivalences, and (ii) an eight-item minimum reportable stack with an auditor and a diff tool that operationalise the lesson.

2 Setting and Protocol

All runs use verifiable rewards: GSM8K exact match (Cobbe et al., 2021) except E-2, which uses synthetic $a+b$ arithmetic, on open models from the Qwen (Yang et al., 2024, 2025) and Llama (Grattafiori et al., 2024) families. Where possible, each comparison fixes model, prompts, verifier, decoding configuration, step budget and held-out evaluation, and varies one factor. The unit of inference is the training seed. Paired t-tests across seeds are reported with df; E-4 follows its preregistered analysis (paired seed bootstrap 95% intervals, Benjamini–Hochberg (Benjamini and Hochberg, 1995) over its four contrasts, exact 80%-power MDE and a ± 0.010 equivalence margin). Exhibits with one or three seeds per arm are descriptive. An inconclusive test is reported as inconclusive; it is not read as equivalence. Where we use the zero-variance fraction (ZVF), it is the

share of a training batch's (post-filtering) prompt groups in which every completion received the same outcome reward, so the mean-centred group-relative advantage is identically zero (with an ϵ-stabilised or omitted standard-deviation divisor).

3 Exhibits

Table 1 summarises the exhibits.

Withdrawn historical exhibits (E-1, E-3). Earlier drafts reported marginal one-way η^2 values over a 42-experiment corpus (E-1) and cross-backend PPO/GRPO statistics for Qwen3-8B and Llama-3.1-8B-Instruct (E-3). No per-run table or analysis script for E-1 survives, and the E-3 statistics were step-level summaries of single runs whose source aggregates cannot be reproduced; recomputing from the retained step logs gives different values. We therefore report neither, and make no direction-reversal or framework-attribution claim.

Matched inconclusive contrast (E-2). With the stack held fixed (one minimal trainer, Qwen2.5-0.5B, synthetic $a{+}b$ arithmetic, 40 steps, five seeds), PPO and GRPO differ by -0.002 in held-out accuracy (GRPO 0.990, PPO 0.992; paired $t_4 = -1.0$, $p = 0.374$): a failure to detect a difference, not equivalence, near a 0.99 ceiling. The stored runs drew evaluation items from a shared random stream, so the two arms of a seed saw different items and PPO's items overlapped its training draws more often; the pairing is by seed only.

Same-stack variants do not separate (E-4). Holding one stack fixed (TRL GRPOTrainer, Qwen3-8B, GSM8K, $G = 4$, 30 steps), we compared DAPO, GSPO, Dr. GRPO and AERO (Zhang et al., 2026) against GRPO over eight paired seeds, evaluating each final checkpoint on 500 GSM8K test items (GRPO mean 0.633). All four contrasts are inconclusive, e.g. GSPO $+0.00500$ (bootstrap 95% CI $[-0.00125, +0.01200]$, raw paired t_7 $p = 0.210$) and AERO -0.00075 ($[-0.00825, +0.00675]$, $p = 0.858$); Benjamini–Hochberg adjustment rejects none. Our own aggregation script had earlier labelled the differences "disappear"; we retract that verdict because each contrast's exact 80%-power MDE (0.0101 for DAPO, up to 0.0148 for Dr. GRPO) exceeds the preregistered ± 0.010 margin, which the preregistered equivalence rule requires. No arm produced a significant held-out difference, and none met that rule. The runs also deviate from parts of the preregistered run contract, so this is a reanalysis of finished runs rather than a validated confirmatory audit.

Labels hide stacks (E-5, E-6). Under the GRPO label, last-10 training reward was 85.6% on the managed service (Qwen3-8B-Base) and 5.0% with TRL on an H100 (Qwen3-8B), one seed and 30 steps each. We do not report this as a backend effect, because the run configurations bind different base checkpoints; that bundling is itself the finding. Similarly, the label "DAPO" produced mean ZVF 0.00 with TRL's dynamic sampling (Qwen3-8B, eight seeds of E-4) and 0.58 on the managed service (Qwen3.5-4B, three seeds, 15 steps), where our own loss under that label clips the standardised advantage to $[-2, 3]$ and has neither an importance ratio nor dynamic sampling. Loss, model and budget all differ, so the telemetry gap cannot be attributed to a single mechanism.

Labels on one managed stack at short horizons (E-7). Running losses labelled GRPO, DAPO, Dr. GRPO and GSPO on one managed stack (Qwen3.5-4B, GSM8K, $G = 8$, 15 steps, three seeds per arm), mean last-10 training reward spans 0.710–0.744, a band of 0.034, smaller than the within-label seed SD (0.090–0.116); the label term gives one-way $\eta^2 = 0.025$ over the 12 runs. These losses are our own simplified implementations (e.g. the GSPO label has no sequence-level ratio), so E-7 shows labels that do not separate, not a comparison of the published algorithms. An earlier group-size variance comparison is withdrawn.

4 A Minimum Reportable Stack

The exhibits and complementary reporting practice motivate eight proposed items: (1) the loss form (ratio level, clip bounds, token mask, advantage normalisation, dynamic sampling), since E-6 shows one label with and without dynamic sampling, and E-7 shows labels attached to simplified losses; (2) reference policy and KL handling; (3) sampler, backend and precision, including base-checkpoint identity, motivated by E-5; (4) the per-step ZVF and gradient-utilisation trajectory, since a single mean conflates opposite regimes; (5) the group-size schedule; (6) a held-out split disjoint from the reward environment; (7) a contamination check and parser-robustness probe; and (8) held-out pass@k curves attached

Exhibit	Result	Reading
E-2 Same-stack PPO vs. GRPO (Qwen2.5-0.5B, arithmetic, 5 paired seeds)	Paired held-out $\Delta = -0.002$, $t_4 = -1.0$, $p = 0.374$	Failure to detect, not equivalence; eval items not matched across arms
E-4 Same-stack variant audit (Qwen3-8B, GSM8K, 8 paired seeds)	DAPO $+0.00100$, GSPO $+0.00500$, Dr. GRPO -0.00200, AERO -0.00075 vs. GRPO; all inconclusive	Per-contrast MDE 0.0101–0.0148 exceeds the 0.010 margin
E-5 Same label, backend/checkpoint bundle (1 seed each)	Last-10 training reward 5.0% to 85.6% ($\approx 17\times$)	Stacks also differ in base checkpoint; not a backend-only effect
E-6 Same label "DAPO", two stacks	Mean ZVF 0.00 (TRL, dynamic sampling) vs. 0.58 (managed stack, our advantage-clip loss)	One label, different losses, models and budgets
E-7 Four labels, one managed stack (4×3 seeds)	Last-10 reward 0.744 DAPO, 0.742 Dr. GRPO, 0.723 GRPO, 0.710 GSPO; band 0.034	Our simplified losses; descriptive, not equivalence

Table 1: Stack-sensitivity exhibits. Rewards are training or held-out accuracy as stated in the text. Historical E-1 and E-3 are withdrawn (see text).

to a standalone held-out evaluation record, not merely in-training rollout telemetry. The items extend general reporting norms (Pineau et al., 2021; Mitchell et al., 2019; Gebru et al., 2021) to the specifics of group-relative RL.

The current implementation is in `platform_hybrid/registry/` of an anonymised repository (withheld for review): `schema.json`, `entries/` and `query.py`. This is a repository locator, not a frozen release or proof of historical execution. Stack records describe configurations; variant-delta records describe declared changes, not new runs. The current badge averages substantive non-null leaf coverage equally over seven schema blocks, then rounds 100 times that mean. Explicit false counts as reported; disabled clipping excludes the two clip-bound leaves. The blocks cover loss form (ratio, clip bounds, length/advantage normalisation, mask, aggregation, shaping, filter, clipping flag), reference/KL (policy, coefficient, estimator), sampler (backend, precision, temperature, top-p), telemetry (ZVF, GU), group schedule (initial G, schedule, rule), held-out split (disjointness, n), and decontamination (performed, parser probe). Pass@k is the eighth recommendation, not an eighth scored block.

Current stackdiff assigns the maximum severity of known differences: R0 no known differences; R1 seed/version/reporting; R2 runtime; R3 hyperparameters/budget; R4 loss/KL/variant changes; R5 outcome-critical checkpoint/reward/prompt/data/evaluation/execution differences. Unknown fields lower coverage and flag underspecification; they do not automatically raise severity. Thus R0 is not proof of a matched stack. We withdraw the obsolete null-implies-R5 interpretation and historical 43–96 badge claim as a description of this current implementation.

Proposal versus compliance. The eight items are a proposed reporting target, not a claim that all historical exhibits satisfy it. Reference/KL disclosure, dynamic schedules, contamination/parser probes and held-out pass@k are complementary reporting recommendations rather than experimentally validated levers in every exhibit. Several historical runs lack checkpoint bindings, trajectories, independent held-out evaluations or contamination evidence. Initial registry manifests are not independent training seeds, and neither their count nor a reporting badge validates scientific quality or hidden implementation semantics.

5 Discussion

None of these exhibits shows that the GRPO variants are equivalent. They show that, at the scale and power available to us, the comparisons that are reported here could not have distinguished a method effect from a stack effect. We think the productive response is procedural rather than algorithmic: report the stack, audit it mechanically, and treat a label-only comparison as unfalsifiable.

Inferential boundary. No direction-reversal or framework-attribution claim is made; E-1 and E-3 are withdrawn. E-5 bundles checkpoint changes and is single-seed. E-7 compares simplified losses under published labels; its narrow training-reward band establishes neither equivalence nor a variance decomposition. These short-horizon (15–40 step), low-seed exhibits cannot test the long-

horizon dynamics that DAPO, Dr. GRPO and GSPO target.

Limitations

Our evidence comes from small-to-mid open models (0.5B–8B in the reported exhibits), mathematics-style verifiable rewards and short training horizons (15–40 steps in the exhibits), so it says nothing about asymptotic behaviour, preference-model rewards or frontier-scale training. Several comparisons are single-seed or low-seed. The managed service's sampler and optimiser internals are closed, so results on it describe that platform plus our loss code rather than an abstract algorithm. The 5.0%–85.6% exhibit mixes backend and checkpoint changes. The historical twenty-record, ten-fingerprint and 23-sub-field summaries are not revalidated against the current schema and are not current registry statistics. Manifest coverage is not experimental independence or outcome validation. The standard's eight items are motivated by our exhibits and complementary reporting practice and have not been validated by independent groups.

Ethics Statement

This work uses public models and public mathematics benchmarks and involves no human subjects or personal data. Its main risk is over-reading: our nulls are underpowered and should not be cited as evidence that the methods are equivalent.

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